Integration by Parts

ANSWER KEY

Section 1: Standard Products (Polynomial × Exponential / Trig)

1 State the formula for integration by parts.

a. $\int u\,dv = uv - \int v\,du$ ✓

b. $\int u\,dv = uv + \int v\,du$

c. $\int u\,dv = u'v - uv'$

d. $\int u\,dv = \frac{u}{v} - \int u'v$

Derived from the product rule: $\frac{d}{dx}(uv) = u'v + uv' \Rightarrow \int u\,dv = uv - \int v\,du$.

2 Evaluate $\int xe^{2x}\,dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ ✓

b. $\frac{1}{2}xe^{2x} - \frac{1}{2}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{4}xe^{2x} - \frac{1}{4}e^{2x} + C$

$u = x, dv = e^{2x}dx \Rightarrow du = dx, v = \frac{1}{2}e^{2x}$. $\int xe^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{2}\int e^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$.

3 Evaluate $\int x\sin(x)\,dx$.

a. $-x\cos x + \sin x + C$ ✓

b. $x\cos x - \sin x + C$

c. $-x\cos x - \sin x + C$

d. $x\sin x + \cos x + C$

$u = x, dv = \sin x dx \Rightarrow du = dx, v = -\cos x$.
$uv - \int v\,du = -x\cos x - \int(-\cos x)dx = -x\cos x + \sin x + C$.

Section 2: Logarithmic and Inverse Trig Integrands

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4 Evaluate $\int \ln(x)\,dx$.

a. $x\ln x - x + C$ ✓

b. $\frac{1}{x} + C$

c. $x\ln x + x + C$

d. $\frac{(\ln x)^2}{2} + C$

$u = \ln x, dv = dx \Rightarrow du = \frac{1}{x}dx, v = x.\ x\ln x - \int x \cdot \frac{1}{x}dx = x\ln x - x + C.$

5 Evaluate $\int \arctan(x)\,dx$.

a. $x\arctan x - \frac{1}{2}\ln(1+x^2) + C$ ✓

b. $x\arctan x - \ln(1+x^2) + C$

c. $\frac{1}{1+x^2} + C$

d. $x\arctan x + \frac{1}{2}\ln(1+x^2) + C$

$u = \arctan x, dv = dx \Rightarrow du = \frac{dx}{1+x^2}, v = x.\ x\arctan x - \int \frac{x}{1+x^2}dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C.$

6 Evaluate $\int x\ln(x)\,dx$.

a. $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$ ✓

b. $\frac{x^2}{2}\ln x - \frac{x^2}{2} + C$

c. $x^2\ln x - x^2 + C$

d. $\frac{x^2}{2}\ln x + \frac{x^2}{4} + C$

$u = \ln x, dv = xdx \Rightarrow du = \frac{dx}{x}, v = \frac{x^2}{2}.\ \frac{x^2}{2}\ln x - \int \frac{x}{2}dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C.$

Section 3: Repeated Integration and Tabular Method

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7 Evaluate $\int x^2 e^x \, dx$.

a. $x^2e^x - 2xe^x + 2e^x + C$ ✓

b. $x^2e^x - 2xe^x + C$

c. $x^2e^x + 2xe^x + 2e^x + C$

d. $\frac{x^3}{3}e^x + C$

Tabular signs $(+, -, +)$: $(+)(x^2)(e^x) - (2x)(e^x) + (2)(e^x) = e^x(x^2 - 2x + 2) + C$.

8 Evaluate $\int x^2 \cos(x) \, dx$.

a. $x^2\sin x + 2x\cos x - 2\sin x + C$ ✓

b. $x^2\sin x - 2x\cos x + 2\sin x + C$

c. $-x^2\cos x + 2x\sin x + C$

d. $x^2\sin x + 2x\cos x + C$

Tabular: $D: x^2, 2x, 2, 0$; $I: \cos x, \sin x, -\cos x, -\sin x$. Pair with signs: $x^2\sin x - 2x(-\cos x) + 2(-\sin x) = x^2\sin x + 2x\cos x - 2\sin x + C$.

9 Evaluate $\int_0^1 x^2 e^{2x} \, dx$.

a. $\frac{e^2 - 1}{4}$ ✓

b. $\frac{e^2}{4}$

c. $\frac{e^2 + 1}{4}$

d. $e^2 - 1$

Tabular antiderivative: $e^{2x}\left(\frac{x^2}{2} - \frac{2x}{4} + \frac{2}{8}\right) = e^{2x}\left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{4}\right)$. At $x = 1$: $e^2(1/2 - 1/2 + 1/4) = \frac{e^2}{4}$. At $x = 0$: $1(1/4) = 1/4$. Difference $= \frac{e^2 - 1}{4}$.

Section 4: Looping / Cyclic Integrals and Definite Bounds

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10 Evaluate $\int e^x \cos(x)\,dx$.

a. $\frac{1}{2}e^x(\cos x + \sin x) + C$ ✓

b. $\frac{1}{2}e^x(\cos x - \sin x) + C$

c. $e^x(\sin x + \cos x) + C$

d. $\frac{1}{2}e^x \sin x + C$

Let $I = \int e^x \cos x\,dx$. By parts twice:
$I = e^x \sin x + e^x \cos x - I \Rightarrow 2I = e^x(\sin x + \cos x) \Rightarrow I = \frac{1}{2}e^x(\sin x + \cos x) + C$.

11 Evaluate $\int_1^e \ln(x)\,dx$.

a. 1 ✓

b. $e - 1$

c. e

d. 0

$[x\ln x - x]_1^e = (e\ln e - e) - (1\ln 1 - 1) = (e - e) - (0 - 1) = 0 - (-1) = 1$.

12 Evaluate $\int_0^{\pi} x\sin(2x)\,dx$.

a. $-\pi/2$ ✓

b. $\pi/2$

c. 0

d. $-\pi$

Parts: $u = x$. $dv = \sin(2x)dx \Rightarrow v = -\frac{1}{2}\cos(2x)$.
$[-\frac{1}{2}x\cos(2x)]_0^{\pi} + \frac{1}{2}\int_0^{\pi}\cos(2x)dx = [-\frac{1}{2}\pi(1) - 0] + [\frac{1}{4}\sin(2x)]_0^{\pi} = -\frac{\pi}{2} + 0 = -\frac{\pi}{2}$.

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